Session 1 (2h) — Sketch Fundamentals & Constraints

Why sketches matter

Every 3D part in Fusion 360 starts as a 2D **sketch** on a plane (XY/XZ/YZ or a flat face). A messy, under-constrained sketch causes every downstream feature (extrude, fillet, pattern) to break when you edit it later — so this session is entirely about building sketches that don't fall apart.

Core concepts (1h)

- **Planes & origin** — every new sketch needs a plane to sit on. Start on the origin's XY plane until you're comfortable.
- **Basic sketch tools** — line, rectangle, circle, arc, polygon.
- **Constraints** (the important part):
  - Geometric: coincident, parallel, perpendicular, tangent, equal, symmetric
  - Dimensional: distance, angle, radius/diameter
- **Fully constrained vs under-constrained** — Fusion turns sketch lines **black** when fully constrained, and leaves them **blue** when something can still move. On-site work expects fully constrained sketches (so a teammate can safely edit your dimensions later).
- **Sketch dimensions vs. drawing to scale** — you draw roughly, then use dimensions to lock the exact size. Never try to draw "by eye" to the correct size.

Hands-on practice (1h)

1. New sketch on the XY plane.
2. Draw a rectangle, then fully constrain it with two dimensions (width, height) + a coincident constraint pinning one corner to the origin.
3. Add a circle inside the rectangle; constrain its center with distance dimensions from two edges, and its size with a diameter dimension.
4. Confirm the whole sketch turns black (fully constrained) — if any line stays blue, find and fix the missing constraint before moving on.
5. Repeat once more with a hexagon (polygon tool) — practice the "equal" and "symmetric" constraints so it doesn't deform.

Checkpoint

- [ ] Can explain the difference between a geometric and a dimensional constraint
- [ ] Produced one fully constrained sketch (all-black geometry) with a rectangle + circle
- [ ] Understands why a real internship team would reject an under-constrained sketch